

XIV. Walsh Transform (Hadamard Transform)

© 14-A Ideas of Walsh Transforms

- 8-point Walsh transform

low frequency ↑ high frequency ↓	$W[m,n] =$	$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 \end{bmatrix}$	m	zero crossings (sign changes)
			0	0 DC term
			1	1
			2	2
			3	3
			4	4
			5	5
			6	6
			7	7

- Advantages of the Walsh transform:

- {

(1) Real 乘法量=0

(2) No multiplication is required

(3) Some properties are similar to those of the DFT

- Forward and inverse Walsh transforms are similar.

$$\text{forward: } F[m] = \sum_{n=0}^{N-1} f[n]W[m,n], \quad \text{inverse: } f[m] = \frac{1}{N} \sum_{n=0}^{N-1} W[m,n]F[n]$$

- Alternative names of the Walsh transform:

Hadamard transform, Walsh-Hadamard transform

- Orthogonal Property $\sum_{n=0}^{N-1} W[m_0,n]W[m_1,n] = 0$ if $m_0 \neq m_1$ $\sum_{n=0}^{N-1} W[m,n]W[m,n] = N$
- Zero-Crossing Property
- Even / Odd Property
- **Fast Algorithm**

Useful for spectrum analysis

Sometimes also useful for implementing the convolution

Walsh transform 和 DFT, DCT 有許多相似處

$$W[m, n] = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix}, DFT[m, n] = \exp(-j2\pi m n/N),$$

8 point DCT multiplied by $\sqrt{8}$
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$$\text{DCT} = \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 & 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.3870 & 1.1759 & 0.7857 & 0.2759 & -0.2759 & -0.7857 & -1.1759 & -1.3870 \\ 1.3066 & 0.5412 & -0.5412 & -1.3066 & -1.3066 & -0.5412 & 0.5412 & 1.3066 \\ 1.1759 & -0.2759 & -1.3870 & -0.7857 & 0.7857 & 1.3870 & 0.2759 & -1.1759 \\ 1.0000 & -1.0000 & -1.0000 & 1.0000 & 1.0000 & -1.0000 & -1.0000 & 1.0000 \\ 0.7857 & -1.3870 & 0.2759 & 1.1759 & -1.1759 & -0.2759 & 1.3870 & -0.7857 \\ 0.5412 & -1.3066 & 1.3066 & -0.5412 & -0.5412 & 1.3066 & -1.3066 & 0.5412 \\ 0.2759 & -0.7857 & 1.1759 & -1.3870 & 1.3870 & -1.1759 & 0.7857 & -0.2759 \end{bmatrix}$$

zero crossings

0

1

2

3

4

5

6

7

References for Walsh Transforms

- [1] K. G. Beanchamp, *Walsh Functions and Their Applications*, Academic Press, New York, 1975.
- [2] B. I. Golubov, A. Efimov, and V. Skvortsov, *Walsh Series and Transforms: Theory and Applications*, Kluwer Academic Publishers, Boston, 1991.
- [3] H. F. Harmuth, “Applications of Walsh functions in communications,” *IEEE Spectrum*, vol. 6, no. 11, pp. 82-91, Nov. 1969.
- [4] H. F. Harmuth, *Transmission of Information by Orthogonal Functions*, Springer-Verlag, New York, 1972.

◎ 14-B Generate the Walsh Transform

2-point Walsh transform

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

4-point Walsh transform

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix}$$

How do we obtain the 2^{k+1} -point Walsh transform from the 2^k -point Walsh transform ?

Step 1

$$\mathbf{V}_{2^{k+1}} = \begin{bmatrix} \mathbf{W}_{2^k} & \mathbf{W}_{2^k} \\ \mathbf{W}_{2^k} & -\mathbf{W}_{2^k} \end{bmatrix}$$

Step 2 根據 sign changes 將 rows 的順序重新排列

$$\mathbf{V}_{2^{k+1}} \xrightarrow{\text{permutation}} \mathbf{W}_{2^{k+1}}$$

已知 W_{2^k} 每個 row 的 sign change 數，由上到下分別為

$$0, 1, 2, 3, \dots, 2^k - 1$$

則 $V_{2^{k+1}}$ 每個 row 的 sign change 數，由上到下分別為

$$0, 3, 4, 7, \dots, 2^{k+1} - 1, 1, 2, 5, 6, \dots, 2^{k+1} - 2,$$

若 row 的 index 由 0 開始

則 $V_{2^{k+1}}$ 第 n 個 row (n is even and $n < N/2$) 的 sign change 為 $2n$

(n is odd and $n < N/2$) 的 sign change 為 $2n + 1$

(n is even and $n \geq N/2$) 的 sign change 為 $2n + 1 - N$

(n is odd and $n \geq N/2$) 的 sign change 為 $2n - N$

要根據 sign change 的數目將 $V_{2^{k+1}}$ 的 row 重新排列

$$V_{2^{k+1}} \xrightarrow{\text{permutation}} W_{2^{k+1}}$$

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\mathbf{V}_4 = \begin{bmatrix} \mathbf{W}_2 & \mathbf{W}_2 \\ \mathbf{W}_2 & -\mathbf{W}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

sign changes

0
3
1
2

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$\mathbf{V}_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \end{bmatrix}$$

sign changes

0
3
4
7
1
2
5
6

Constraint for the number of points of the Walsh transform:

N must be a power of 2 (2, 4, 8, 16, 32,)

Although in Matlab it is possible to define the $12 \cdot 2^k$ point or the $20 \cdot 2^k$ point Walsh transform, the inverse transform require the floating-point operation.

© 14-C Alternative Forms of the Walsh Transform

- Sequency ordering (i.e., Walsh ordering) using 標準定義 from zero-crossing for signal processing
- Dyadic ordering (i.e., Paley ordering) using for control
- Natural ordering (i.e., Hadamard ordering) using for mathematics

Sequency ordering	Dyadic ordering	Natural ordering	$W[m, n]$
	← (Gray Code) ←	→ (Bit Reversal) →	
row 0 =	000 row 0 =	000 row 0 =	[1, 1, 1, 1, 1, 1, 1, 1]
row 1 =	001 row 1 =	100 row 4 =	[1, 1, 1, 1, -1, -1, -1, -1]
row 2 =	011 row 3 =	110 row 6 =	[1, 1, -1, -1, -1, -1, 1, 1]
011 row 3 =	010 row 2 =	010 row 2 =	[1, 1, -1, -1, 1, 1, -1, -1]
100 row 4 =	110 row 6 =	011 row 3 =	[1, -1, -1, 1, 1, -1, -1, 1]
row 5 =	111 row 7 =	111 row 7 =	[1, -1, -1, 1, -1, 1, 1, -1]
row 6 =	101 row 5 =	101 row 5 =	[1, -1, 1, -1, -1, 1, -1, 1]
row 7 =	100 row 4 =	001 row 1 =	[1, -1, 1, -1, 1, -1, 1, -1]

- Dyadic ordering
Walsh transform

$$W[m,n] = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix}$$

- Natural ordering
Walsh transform

$$W[m,n] = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix}$$

- binary code $n = \sum_{p=1}^k b_p 2^{p-1}$ to gray code
When $N = 2^k$

$$g_k = b_k, \quad g_q = \text{XOR}(b_{q+1}, b_q) \quad \text{for } q = k-1, k-2, \dots, 1 \quad m = \sum_{q=1}^k g_q 2^{q-1}$$

- gray code to binary code

When $N = 2^k$

$$b_k = g_k, \quad b_q = \text{XOR}(b_{q+1}, g_q) \quad \text{for } q = k-1, k-2, \dots, 1$$